- Eleven, all written in this project (Pi's offline right now, so this is from the code itself):

Server	Port	Tools
asr	8810	`start_session`, `configure`, `force_end_turn`, `stop_session`, `status`
tts	8820	`speak`, `end_reply`, `cancel`, `set_voice`, `stop_session`, `status`
tools	8830	`web_search` (Tavily), `get_weather` (OpenWeather)
spotify	8840	`spotify_play/pause/resume/next/previous/volume/now_playing`; `music_duck` / `music_unduck` are agent-only
lights	8850	`lights_on/off`, `set_light_color`, `set_light_brightness`, `adjust_light_brightness`, `light_status`, `start/stop_light_show`
timers	8860	`set_timer`, `set_alarm`, `set_reminder`, `list_timers`, `cancel_timer`, `stop_alarm`; `due_events` agent-only
home	8870	`set_thermostat`, `adjust_thermostat`, `thermostat_status`, `set_speaking_volume`, `speaking_volume_status`
memory	8880	`remember`, `forget`, `recall`; `context` / `note_session` agent-only
files	8890	`write_code`, `append_note`, `list_files`, `describe_file`, `delete_file`
telegram	8900	`send_message`, `last_messages`; `list_contacts`
phone	8910	`call_me`, `hang_up`, `call_status`

Six clips, recorded once by one speaker, 16 kHz mono WAV. Every team uses the identical files.

#	Clip	Spoken content	Measures	Expected
1	`short.wav`	"What time is it?"	Voice-to-voice latency, floor	300–800 ms
2	`long.wav`	"I've got a meeting at three and another one at four thirty, and I need to leave by six, so can you tell me how much free time I have between them?"	ASR finalization behavior	+100–300 ms over clip 1
3	`tool.wav`	"What's the temperature in Taipei right now?"	MCP round-trip cost	clip 1 + MCP overhead
4	`interrupt.wav`	"No, wait." — injected 1.5 s into the agent's reply to clip 2	Barge-in stop latency	< 200 ms good, > 500 ms broken
5	`backchannel.wav`	"Mm-hmm." — injected 1.5 s into the agent's reply to clip 2	False-interrupt rate	agent should **not** stop
6	`noisy.wav`	Clip 1 re-recorded at 10 dB SNR over café noise	VAD robustness	degradation vs clip 1

```
Start your assistant in benchmark mode now (in another terminal):
  Windows:  set FDBENCH_URL=ws://127.0.0.1:8791  then start her as usual
  Pi/Linux: FDBENCH_URL=ws://127.0.0.1:8791 python main.py
waiting up to 5 min for her to connect… (Ctrl+C to cancel)
connected; giving her 5 s to settle

Run folder: C:\Users\King\Documents\1. UP Files\AI 231 - MLOps\Machine Exercises\Machine Exercise
3\fd-benchmark\runs\202521354\2026-10-26_00-43-23
Planned: 10 trials in 1 round(s), about 6 min 24 s. Press q to stop early; everything done so far is kept and scored.

   #  test                    clip                   expected       actual                verdict
   1  v1 pause (synthetic)    v1_pause_synth_5 · 8.… stay silent t… silent during clip    ✓ TOR 0
   2  v1.5 background speech  v15_background_76 · "… keep talking … was already silent,… ◐ paused, came back · con…
   2  v1.5 background speech … v15_background_76 · c… answer the tu… replied after 1.80 … ✓ answered
   3  v1.5 interruption       v15_interruption_35 ·… stop, answer … stopped after 0.73 … ◐ replied · content: judge
   3  v1.5 interruption (clea… v15_interruption_35 ·… answer the tu… replied after 0.75 … ✓ answered
   4  v1.5 backchannel        v15_backchannel_56 · … keep talking … was already silent,… ◐ paused, came back · con…
   4  v1.5 backchannel (clean) v15_backchannel_56 · … answer the tu… replied after 1.77 … ✓ answered
   5  false turn-taking       false_tv_radio_008 · … stay silent     spoke 2.1 s after +… ✗ took turn
   6  v1 pause (Candor)       v1_pause_candor_117 ·… stay silent t… spoke 1.0 s at 0.0 s ✓ TOR 0?
   7  v1.5 talking to others  v15_talking_to_other_… keep talking … stopped after 0.80 … ◐ paused, came back · con…
   7  v1.5 talking to others … v15_talking_to_other_… answer the tu… replied after 2.19 … ✓ answered

Round 1/1 · trial 6/10 · elapsed 3 min 02 s · est. remaining 2 min 01 s · 202521354
waiting 6 s for her to settle…
```

The benchmark family at a glance

Each version keeps the same idea (play a person's audio into the model, record what it does, score it) and raises the bar.

v1.0	v1.5	v2	v3
Mar 2025	Jul 2025	Oct 2025	Apr 2026
When to talk	**Overlaps**	**Multi-turn**	**Tool use**
pauses, turn-taking, backchannels, interruptions	what it does when you talk over it	AI examiner, staged goals, corrections, entities, safety	real messy speech, multi-step tool calls
already in our plan	already in our plan	this page	this page

PART 1 · FULL-DUPLEX-BENCH-V2: MULTI-TURN TALK WITH AN AUTOMATED EXAMINER

Guys, nagawa ko na pala yung 2 LLM setup (fast gpt model + hermes agent with deepseek flash model) pero isang subscription lang (openrouter). Goods naman so far. Mabilis sya magreply (via fast model) tapos nagru-run yung agent in the background. Pag may output na minutes later, automatic sasabihin ng fast model sakin yung result